Self-Attention on Computer Vision

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Introduction

1. Self-attention has made great progress on sequence modeling.

2. Context is crucial for many vision task, such as semantic segmentation, object detection.

3. Self-attention provides an useful and efficient way for context modeling.

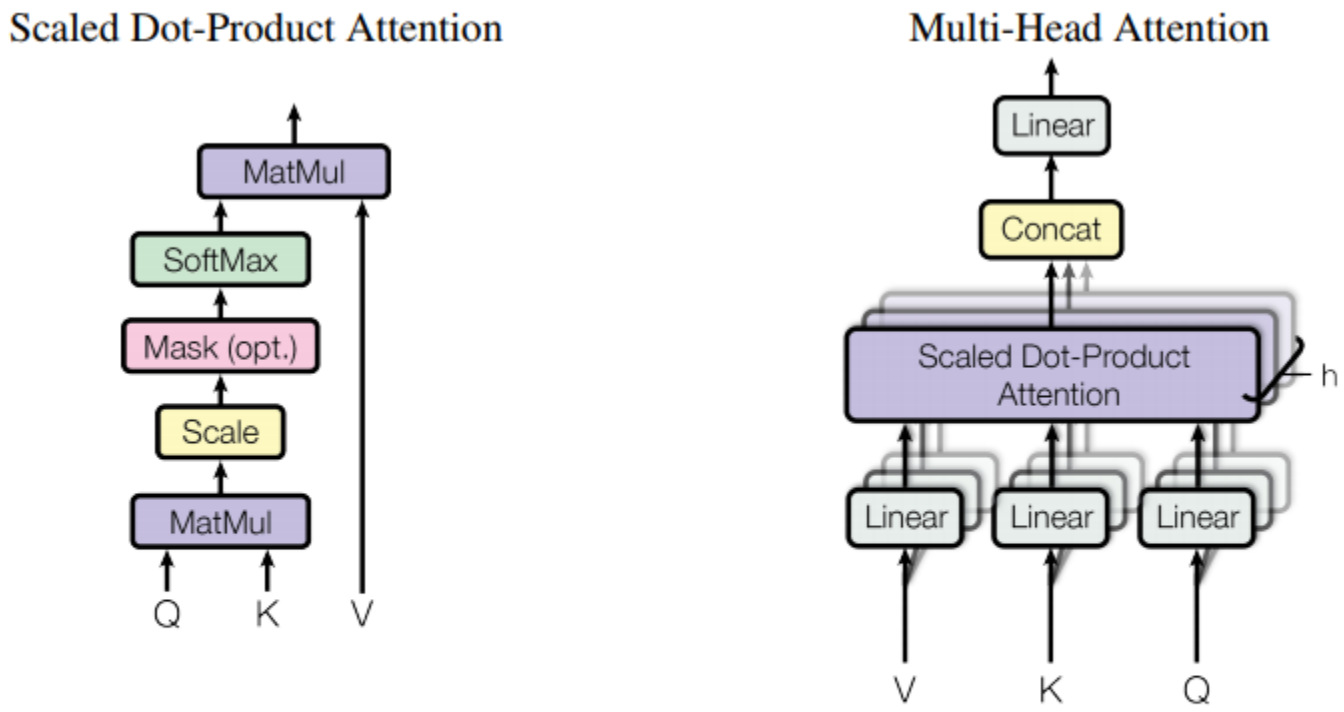


Figure 2: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser and Illia Polosukhin. Attention Is All You Need. arXiv:1706.03762, 2017

Google的大作提供了**第三个思路：纯 Attention，单靠注意力就可以。**

RNN 要逐步递归才能获得全局信息，因此一般要双向 RNN 才比较好；CNN 事实上只能获取局部信息，是通过层叠来增大感受野；Attention 的思路最为粗暴，它一步到位获取了全局信息，它的解决方案是：

$$y_t = f(x_t, A, B)$$

请点击此处输入图片描述

其中 A,B 是另外一个序列（矩阵）。如果都取 A=B=X，那么就称为 Self Attention，**它的意思是直接将 xt 与原来的每个词进行比较，最后算出 yt。**

https://www.jiqizhixin.com/articles/2018-01-10-20

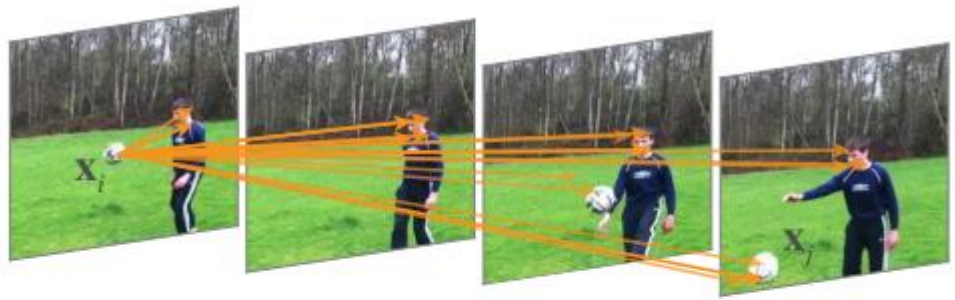


Figure 1. A spacetime ***non-local*** operation in our network trained for video classification in Kinetics. A position $\mathbf{x}_i$'s response is computed by the weighted average of the features of *all* positions $\mathbf{x}_j$ (only the highest weighted ones are shown here). In this example computed by our model, note how it relates the ball in the first frame to the ball in the last two frames. More examples are in Figure 3.

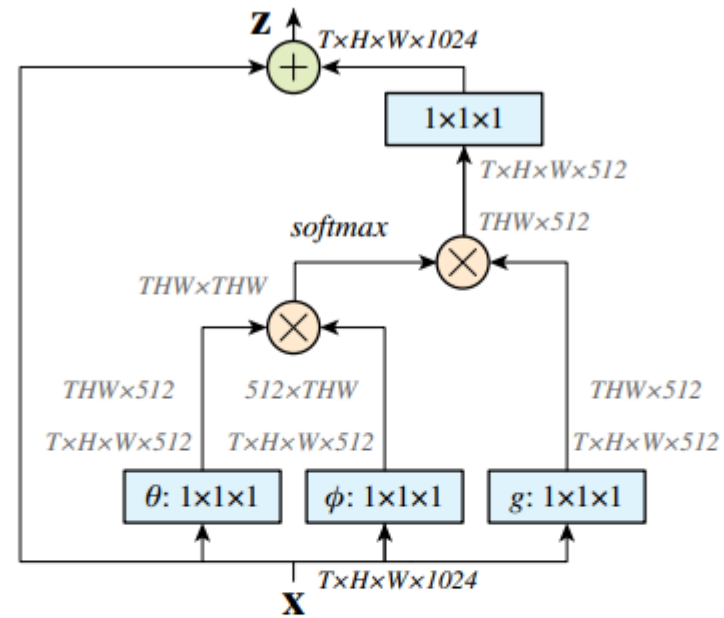
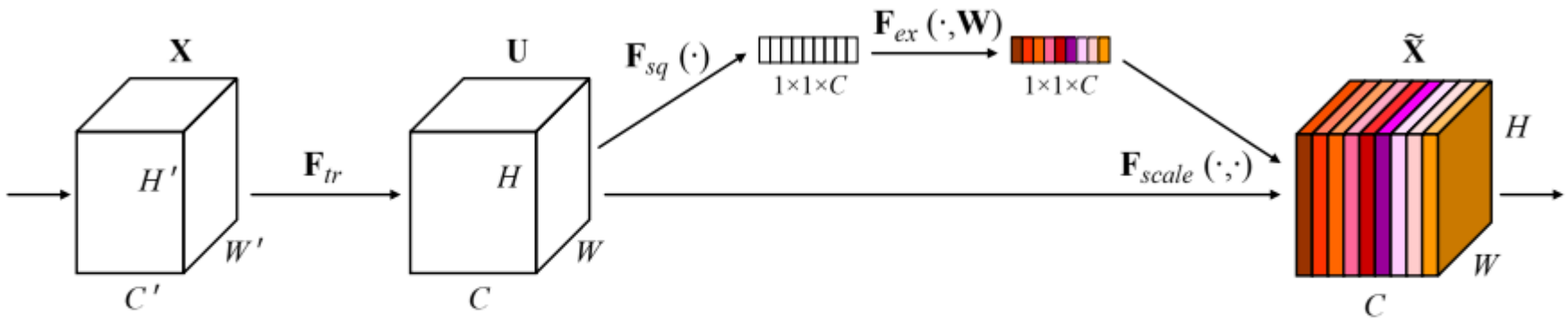
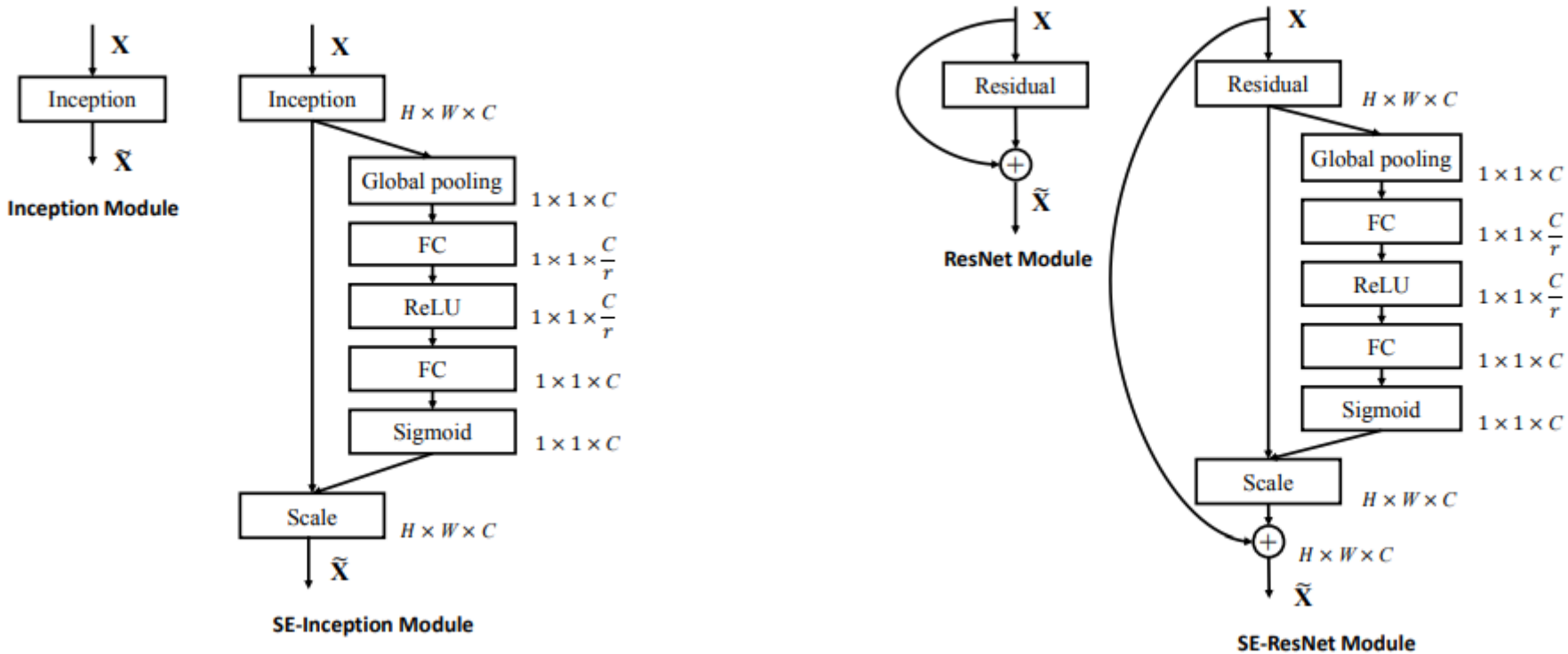


Figure 2. A spacetime **non-local block**. The feature maps are shown as the shape of their tensors, *e.g.*, $T \times H \times W \times 1024$ for 1024 channels (proper reshaping is performed when noted). "$\otimes$" denotes matrix multiplication, and "$\oplus$" denotes element-wise sum. The softmax operation is performed on each row. The blue boxes denote $1 \times 1 \times 1$ convolutions. Here we show the embedded Gaussian version, with a bottleneck of 512 channels. The vanilla Gaussian version can be done by removing θ and ϕ, and the dot-product version can be done by replacing softmax with scaling by $1/N$.

Xiaolong Wang, Ross Girshick, Abhinav Gupta, Kaiming He . Non-local Neural Networks. arXiv:1711.07971, 2017





Jie Hu et al. . Squeeze-and-Excitation Networks. CVPR2017

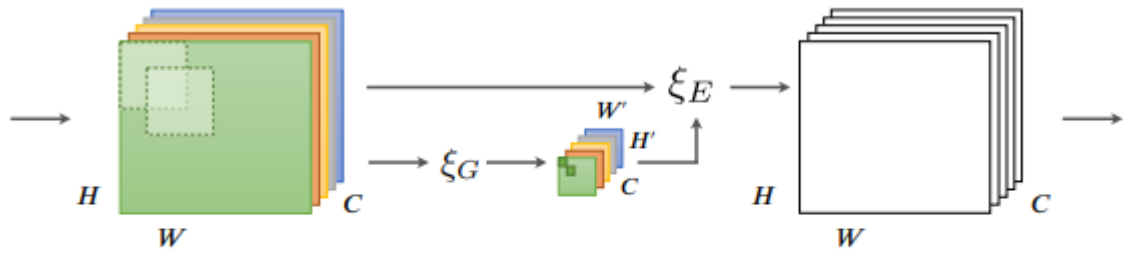
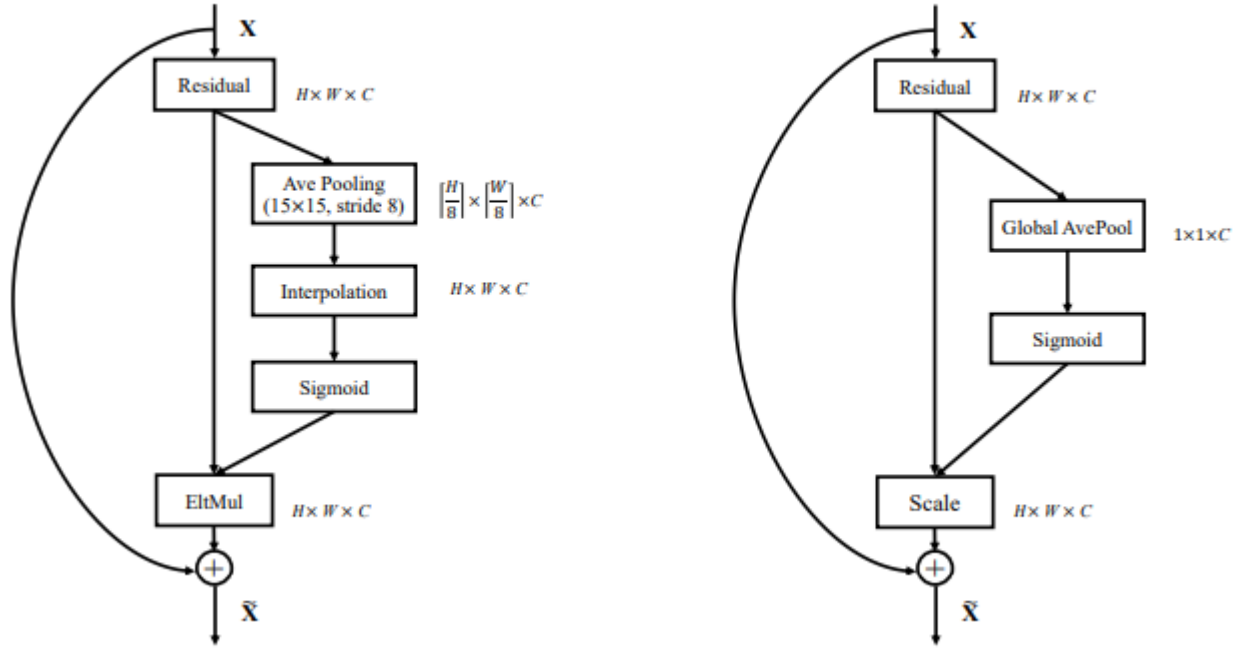


Figure 1: The interaction of a *gather-excite* operator pair, (ξ_G, ξ_E). The gather operator ξ_G first aggregates feature responses across spatial neighbourhoods. The resulting aggregates are then passed, together with the original input tensor, to an excite operator ξ_E that produces an output that matches the dimensions of the input.



Jie Hu et al. . Gather-Excite: Exploiting Feature Context in Convolutional Neural Networks. NIPS2018

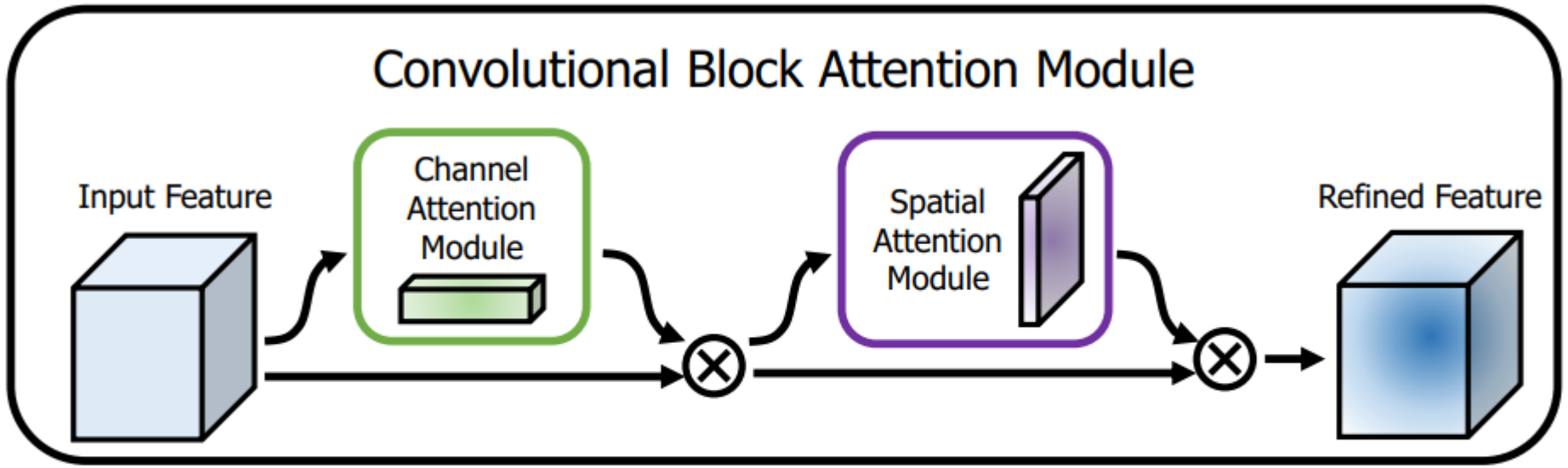


Fig. 1: **The overview of CBAM**. The module has two sequential sub-modules: *channel* and *spatial*. The intermediate feature map is adaptively refined through our module (CBAM) at every convolutional block of deep networks.

Sanghyun Woo et al. . CBAM: Convolutional Block Attention Module . ECCV2018

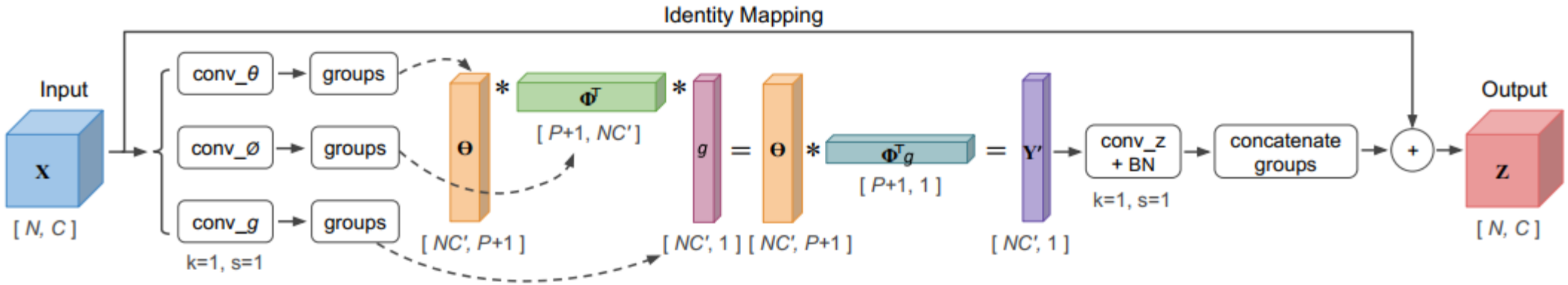


Figure 2: Grouped compact generalized non-local (CGNL) module. The feature maps are shown with the shape of their tensors, *e.g.*, $[C, N]$, where $N = THW$ or $N = HW$. The feature maps will be divided along channels into multiple groups after three conv layers whose kernel size and stride both equals 1 ($\mathrm{k} = 1, \mathrm{s} = 1$). The channels dimension is grouped into $C' = C/G$, where G is a group number. The compact representations for generalized non-local module are build within each group. P indicates the order of Taylor expansion for kernel functions.

Kaiyu Yue et al. Compact Generalized Non-local Network. In NIPS 2018

Application

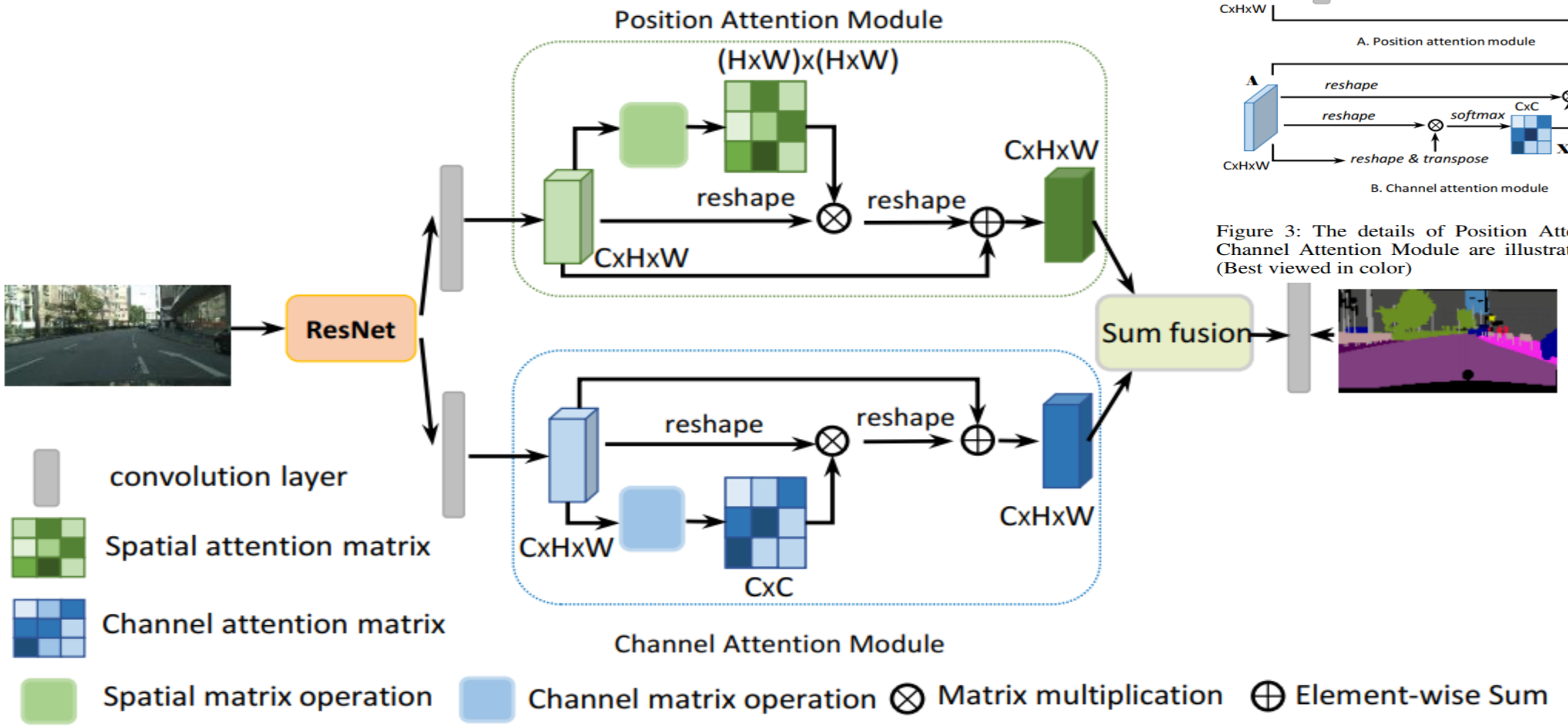


Figure 2: An overview of the Dual Attention Network. (Best viewed in color)

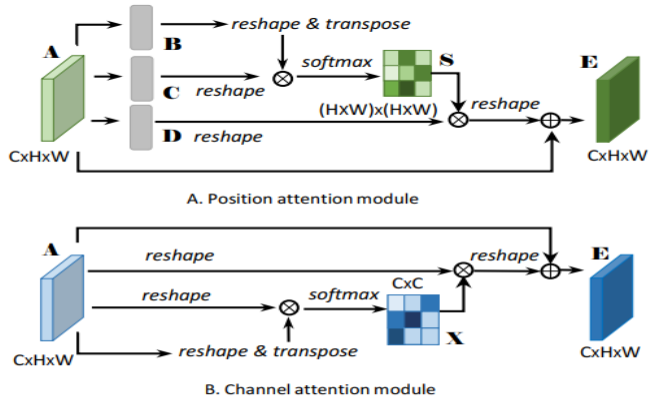


Figure 3: The details of Position Attention Module and Channel Attention Module are illustrated in (A) and (B). (Best viewed in color)

Jun Fu, Jing Liu, Haijie Tian, Zhiwei Fang, Hanqing Lu. Dual Attention Network for Scene Segmentation. arXiv:1809.02983, 2018

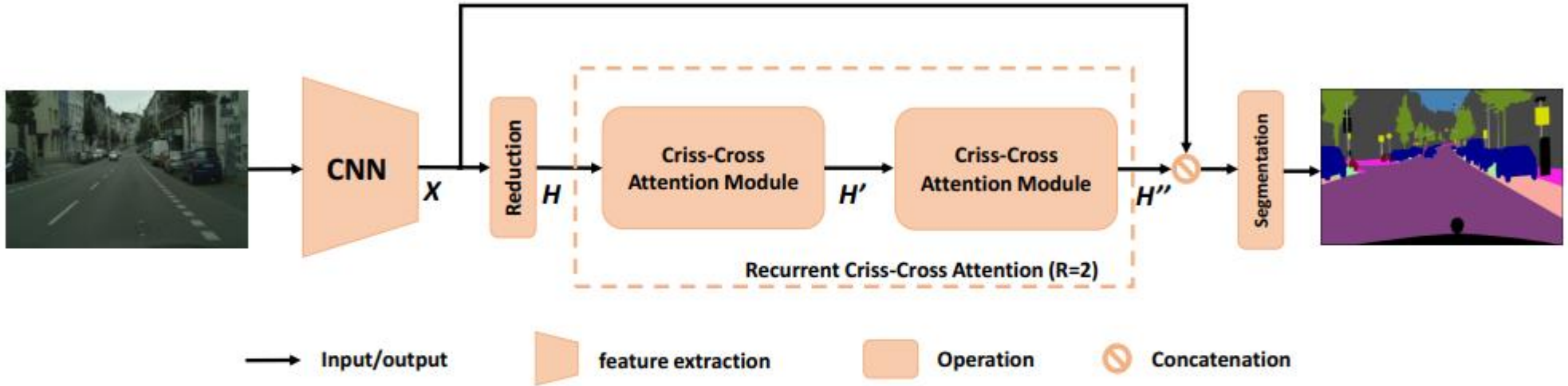
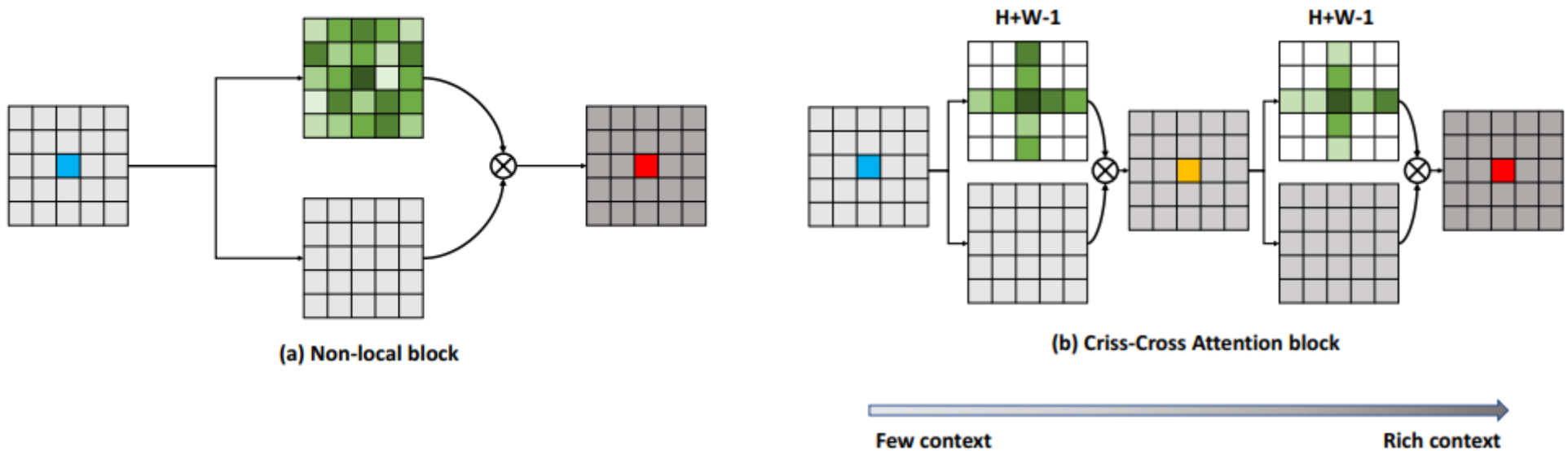


Figure 2. Overview of the proposed CCNet for semantic segmentation. The proposed recurrent criss-cross attention takes as input feature maps $\mathbf{H}$ and output feature maps $\mathbf{H}''$ which obtain rich and dense contextual information from all pixels. Recurrent criss-cross attention module can be unrolled into $R = 2$ loops, in which all Criss-Cross Attention module share parameters.



Zilong Huang et al.. CCNet: Criss-Cross Attention for Semantic Segmentation. arXiv:1811.11721

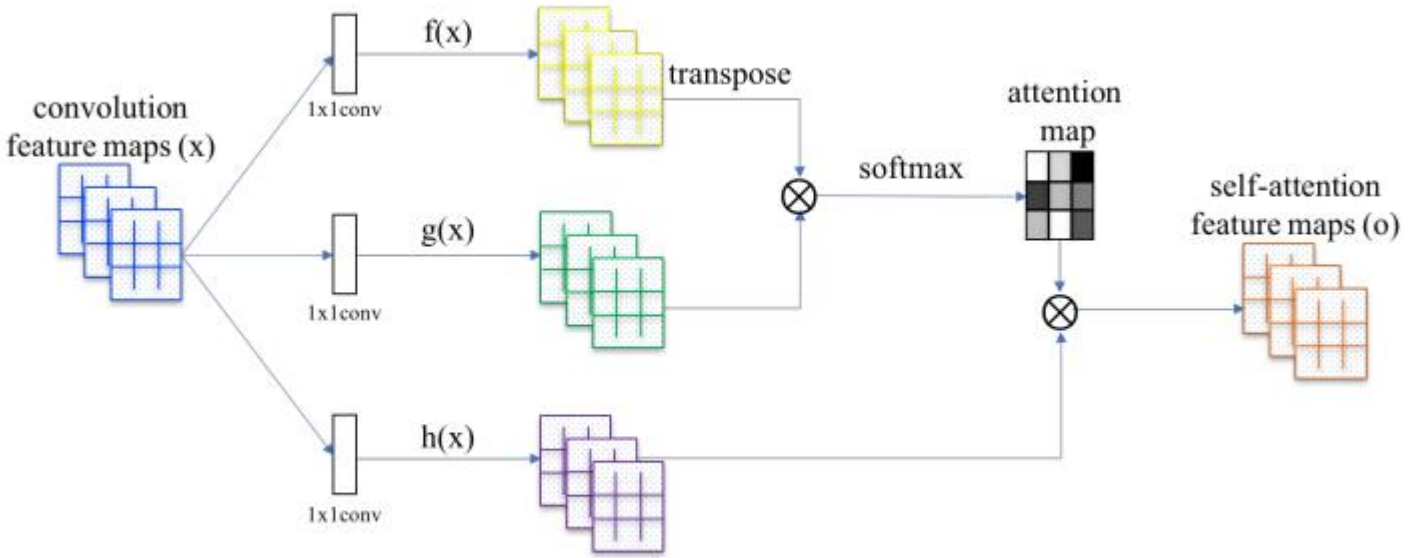


Figure 2: The proposed self-attention mechanism. The ⊗ denotes matrix multiplication. The softmax operation is performed on each row.

SAGAN论文中给出的解释是GAN擅长合成几乎没有结构约束的图像类别，例如海洋、天空和景观类别。但是无法捕获在某些类别中始终如一地出现的几何或结构模式，例如狗的毛皮。这种原因并不是GAN自身带来的，而是以前的GAN模型在很大程度上依赖于卷积来模拟不同图像区域之间的依赖关系。由于卷积运算过程中，是基于一个局部感受域，所以只能在经过几个卷积层之后才能处理图像中远距离的相关性

Han Zhang et al . Self-Attention Generative Adversarial Networks. arXiv:1805.08318, 2018

Rethinking

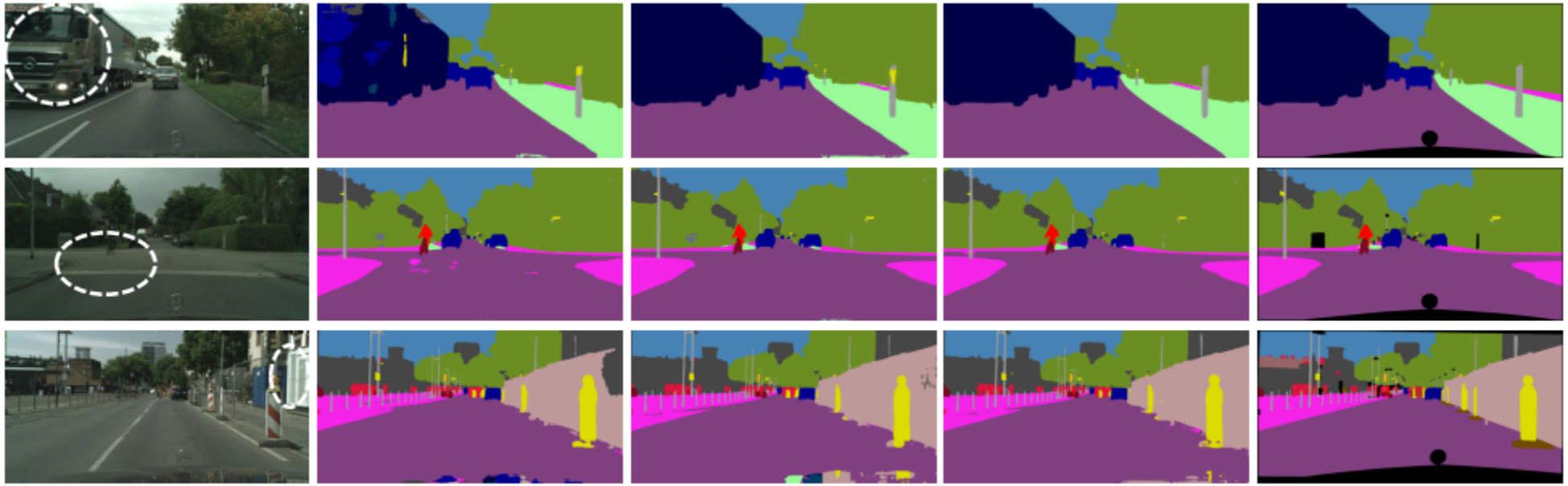


Figure 5. Visualization results of RCCA with different loops on Cityscapes validation set.

How to combine the information of channel and spatial?

- *channel as spatial?*
- *fusion of channel and spatial?*
- *group*

Find the drawbacks of current method and learn from other method.

Table 3. Performance on Cityscapes validation set for different loops in RCCA. FLOPs and Memory usage are estimated for an input of $1 \times 3 \times 769 \times 769$.

Loops	GFLOPs(▲)	Memory(M▲)	mIOU(%)
baseline	0	0	75.1
R=1	8.3	53	78.0
R=2	16.5	127	79.8
R=3	24.7	208	80.2